Sahaj Bhadja

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Summary

Solutions engineer experienced in **Java, Python, SQL, cross-platform troubleshooting, root-cause analysis, customer support, and documentation**, resolving production issues across services, databases, integrations, and distributed systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Computer Networks, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Data Retrieval, Computer Architecture

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Investigated production behavior across FastAPI services, Azure SQL, and background workers using **structured logs, query analysis, explicit job states, and failure isolation**, sustaining 99%+ reliability across 1K+ monthly jobs.
- Optimized multi-tenant data access through **indexed queries, normalized schemas, and resource profiling**, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ daily application users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Troubleshot Flask services, model pipelines, complex browser workflows, and cloud databases supporting 10K+ daily requests, applying **log correlation, reproducible test cases, customer-impact analysis, and documented production resolutions**.
- Developed multithreaded Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing turnaround 60% through **input validation, fault-tolerant processing, integration testing, performance analysis, and release verification**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Diagnosed cross-platform service and data-integration issues across **Windows, Linux, Python, AWS Glue, S3, and SQL**, improving response times 30% through root-cause analysis, query optimization, and durable fixes.
- Automated 20+ unit and integration suites using PyTest, Mockito, Jenkins, and virtual machines, increasing pipeline efficiency 68% through **repeatable environments, regression isolation, diagnostic evidence, and release validation**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Supported eight services for authentication, billing, permissions, resources, and notifications, tracing failures across **REST APIs, WebSockets, Azure SQL, event-driven updates, shared schemas, and reliable real-time application state**.
- Improved production supportability with **Sentry monitoring, structured logging, Docker, Jenkins, and integration tests**, reducing deployment time 50% while documenting recurring defects, release behavior, troubleshooting steps, and corrective actions.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built and supported Flask APIs for bookings, pricing, payments, and real-time status updates across 50K+ records, achieving sub-200-millisecond responses through **SQL analysis, caching, webhook debugging, and defect resolution**.
- Translated customer issues into technical specifications for 12+ integrations, communicating **problem breakdowns, API behavior, data ownership, validation rules, resolution progress, and implementation recommendations** to technical and nontechnical stakeholders.

Skills

Programming & Data: Java, Python, SQL, JavaScript, Bash, MS SQL Server, MySQL, PostgreSQL, Oracle, Database Design
Industrial & Integration: Ignition, SCADA, HMI, PLCs, OPC UA, MQTT, Industrial Networks, REST APIs, WebSockets, IT/OT Integration
Systems & Diagnostics: Windows, Linux, macOS, Command Line, Log Analysis, Root-Cause Analysis, Performance Analysis, Networking, Distributed Systems
Support & Quality: Customer Support, Ticket Management, Knowledge Bases, Technical Documentation, PyTest, Unit Testing, Integration Testing, Git, Docker, Jenkins, CI/CD

Projects

Ignition SCADA Support Lab | *Java, Python, SQL, Ignition, OPC UA, MQTT, Docker*

- Built an **industrial SCADA support environment** connecting simulated PLC tags through OPC UA and MQTT to Ignition, SQL history, alarms, and dashboards across containerized Windows and Linux test configurations.

Industrial Incident Diagnostic Console | *Python, SQL Server, MySQL, PostgreSQL, Linux*

- Developed a **cross-layer diagnostic console** that correlates gateway logs, timestamps, database health, tag quality, network events, and configuration changes, producing reproducible incident timelines, root-cause evidence, and knowledge-base drafts.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven, Node.js* | GitHub

- Extended a cross-platform database workspace supporting 10+ databases through **object-oriented architecture, schema synchronization, validated query execution, containerized environments, and reusable integrations**, simplifying database troubleshooting across heterogeneous customer systems.